fighters, adapt to changing tactical contexts, execute coordinated maneuvers, and accept dynamic tasking with minimal supervisory bandwidth. Key capabilities include autonomous formation keeping, automated collision avoidance, cooperative target handoff, and autonomous execution of delegated missions under pre-approved rules of engagement. Communications and data links are designed for graceful degradation (e.g., operating with intermittent connectivity) and for resilience against electronic attack; embedded AI provides local decision logic when link to the human controller is constrained (Stensrud, et al. 2024).

Operational roles for loyal wingmen are diverse and mission-dependent. In reconnaissance and intelligence roles they can advance ahead of crewed aircraft to map threats and provide standoff sensing. As weapons carriers or decoys they can carry precision munitions or perform suppression of enemy air defense (SEAD) tasks that would otherwise risk human pilots. In contested environments they can act as sensor nodes to extend detection ranges, perform electronic-attack missions to blind adversary sensors, or loiter to provide continuous overwatch while crewed aircraft undertake higher-value tasks. Importantly, loyal wingmen enable task distribution and tempo amplification a single manned aircraft can control a small package of coordinated UAVs to achieve broader situational effect than the manned asset alone.

Despite the operational promise, numerous challenges must be addressed for practical deployment. Ensuring predictable, verifiable autonomy under combat stress is essential to avoid unintended escalation or fratricide; this requires rigorous validation, transparent decision logic, and rehearsed human override mechanisms. Communications must be highly secure and jam-resistant, with contingency modes for degraded connectivity. Logistics mass production, maintenance, rapid rearming, and forward basing pose